DSA0602 Data Handling and Visualization

Slot A

CO1 – Assessment Test 3

Visualization Design Exercise

Instructions

Each student has been randomly assigned 5 different questions (no duplicates) drawn from a common question bank spanning all the topics above.

For each question, describe/sketch the visualization you propose (on paper or as a labelled diagram) and justify your design choices in 3–5 senten

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Question 1 *[Coordinate Systems and Axes — Polar Coordinates]*

A weather station records the average wind direction and speed for each month of the year. Design a Polar-coordinate chart to represent this data, and explain why Polar coordinates suit this dataset better than Cartesian coordinates.

Introduction

Data visualization is the process of presenting information in graphical form to make it easier to understand and interpret. Different types of data require different visualization techniques. Polar coordinates are especially useful when representing data involving angles, directions, and circular patterns.

A weather station records information about wind direction and wind speed throughout the year. Wind direction indicates where the wind comes from, while wind speed indicates how fast the wind moves. Since wind direction is measured in degrees around a circle, a polar-coordinate chart is an appropriate method for visualizing this information.

Design of the Polar-Coordinate Chart

A polar-coordinate chart represents data using two main components: an angle and a distance from the centre. These components can be used to represent wind direction and wind speed.

1. Angle representing wind direction

The angle around the chart represents the direction of the wind. The directions can be marked around the circular chart as follows:

- North: 0° or 360°
- East: 90°
- South: 180°
- West: 270°

Intermediate directions, such as northeast, southeast, southwest, and northwest, can also be marked. This allows the reader to identify the direction from which the wind is blowing.

2. Distance representing wind speed

The distance from the centre represents the average wind speed. A longer distance indicates a higher wind speed, while a shorter distance indicates a lower wind speed.

For example, a wind speed of 10 km/h can be represented by a point closer to the centre than a wind speed of 30 km/h. Concentric circles can be added to the chart to indicate different wind-speed values.

3. Representing the monthly data

Each month can be represented using a point, line, or arrow positioned according to the recorded wind direction and speed. Different colours or symbols can distinguish the months.

For example, if January has an average wind direction of northeast and a speed of 15 km/h, its marker would be placed at the northeast angle and at the distance corresponding to 15 km/h. If July has a wind direction of southwest and a speed of 25 km/h, its marker would be placed at the southwest angle and at the distance corresponding to 25 km/h.

The same method can be used for all twelve months of the year.

4. Adding a legend and labels

The chart should include a suitable title, direction labels, wind-speed measurements, and a legend explaining the symbols or colours used. These elements help readers interpret the information correctly.

Suggested chart title: *Monthly Average Wind Direction and Speed Using Polar Coordinates.*

Why Polar Coordinates Are Better Than Cartesian Coordinates

Polar coordinates are more suitable for this dataset because wind direction is circular in nature. Directions are measured around a complete circle, and the end of the directional scale meets its beginning. For example, north is represented by both 0° and 360°.

In a Cartesian-coordinate chart, information is usually represented using horizontal and vertical axes. Although Cartesian charts are useful for comparing numerical values and identifying relationships between variables, they do not naturally represent circular directions.

A polar-coordinate chart provides the following advantages:

1. Natural representation of direction: Wind direction can be displayed around a circle, making it easy to identify whether the wind comes from the north, south, east, west, or an intermediate direction.

2. Easy comparison of wind speeds: The distance from the centre makes it possible to compare wind speeds. Longer distances indicate stronger winds, while shorter distances indicate weaker winds.

3. Identification of seasonal patterns: By comparing the monthly markers, the reader can identify changes in wind direction and speed across different seasons.

4. Better understanding of circular data: Directions such as 359° and 1° are very close to each other. A polar chart correctly shows that both directions are near north, whereas a simple Cartesian representation of the numerical angles might make them appear far apart.

5. Compact representation: A single polar chart can display wind information for all twelve months, allowing readers to examine the overall pattern without needing twelve separate charts.

Conclusion

Polar coordinates are an effective choice for visualizing monthly average wind direction and speed because they represent direction through angles and speed through distance from the centre. They help identify dominant wind directions, compare wind speeds, and understand seasonal changes. Although Cartesian coordinates are useful for many types of numerical data, polar coordinates provide a more natural and intuitive representation of wind direction.

Question 2 *[Color Scales — Color as Distinction]*

Explain the difference between using colour for distinction (categorical grouping) and using colour for representation (encoding a continuous/quantitative value), with one example of each.

Introduction

Colour is an important visual element in data visualization. It helps readers distinguish categories, identify patterns, compare values, and understand information quickly. However, colour must be used appropriately according to the type of data being represented.

Two common uses of colour in data visualization are colour for distinction and colour for representation. Although both techniques use colours to communicate information, they serve different purposes.

1. Colour for Distinction (Categorical Grouping)

Colour for distinction is used to differentiate between separate categories in a dataset. Each category is assigned a different colour so that readers can easily identify and compare groups.

This technique is particularly useful when working with nominal data, which consists of categories that do not have a natural order.

For example, consider a dataset containing the number of students enrolled in three college departments: Computer Science, Mathematics, and Physics.

A bar chart can use the following colours:

- Blue for Computer Science.
- Green for Mathematics.
- Orange for Physics.

Each colour represents a different department. The colours do not indicate that one department is better, larger, or more important than another. Their main purpose is to distinguish the categories.

A legend should be included to explain what each colour represents.

Example

Suppose a college has the following number of students:

Department	Number of Students	Colour
Computer Science	120	Blue
Mathematics	80	Green
Physics	100	Orange

In this example, colours help readers identify the departments quickly. The height of each bar communicates the number of students, while the colour identifies the department.

Advantages of Colour for Distinction

1. It makes different categories easier to identify.
2. It improves the readability of charts containing multiple groups.
3. It helps readers compare categories without repeatedly reading labels.
4. It makes complex datasets easier to understand.
5. It supports the identification of different groups in charts and diagrams.

When using colour for distinction, the selected colours should be sufficiently different from one another. Using too many similar colours can confuse readers.

2. Colour for Representation (Encoding Quantitative Values)

Colour for representation is used to communicate numerical values or measurements. Different colours or shades indicate different quantities, allowing readers to identify high values, low values, and changes across a dataset.

This technique is useful for continuous or quantitative data, such as temperature, rainfall, population density, and elevation.

A sequential colour scale is commonly used when values range from low to high. For example, light shades may represent low values, while dark shades represent high values.

A diverging colour scale can be used when values are measured relative to a meaningful midpoint, such as zero

or a target value. Different colours can represent values below and above that midpoint.

Example

Consider a weather map showing temperatures recorded in different cities.

City	Temperature	Colour Representation
City A	15°C	Light shade
City B	25°C	Medium shade
City C	35°C	Dark shade

In this example, the colour intensity represents temperature. A darker shade indicates a higher temperature, while a lighter shade indicates a lower temperature.

The colours communicate numerical differences rather than simply identifying separate categories.

A colour legend or colour bar should be provided to show how the shades correspond to temperature values.

Advantages of Colour for Representation

1. It helps readers identify high and low numerical values quickly.
2. It reveals patterns and variations across a dataset.
3. It is useful for visualizing geographical and scientific information.
4. It allows large amounts of numerical data to be displayed compactly.
5. It makes unusual values and areas of concentration easier to identify.

Major Differences Between the Two Uses of Colour

Basis	Colour for Distinction	Colour for Representation
Main purpose	Separates categories	Communicates numerical values
Type of data	Usually categorical or nominal	Quantitative or continuous
Colour selection	Distinct colours for different categories	Ordered shades or colours representing values
Meaning of colour	Identifies a group	Indicates a value or magnitude
Example	Different colours for college departments	Different shades representing temperature
Legend	Explains category-to-colour mapping	Explains colour-to-value mapping
Interpretation	Readers identify which category is shown	Readers estimate or compare numerical values

Conclusion

Colour for distinction and colour for representation are both important techniques in data visualization, but they serve different purposes. Colour for distinction helps readers separate categories, such as departments or product types. Colour for representation communicates numerical values, such as temperature or rainfall, through an ordered colour scale. Choosing the appropriate method improves clarity, accuracy, and the overall effectiveness of a visualization.

Question 3 *[Color Scales — Designing Effective Color Palettes]*

List three principles of effective colour-palette design (e.g., limited hues, consistent meaning across charts, accessibility) and explain each with a short example.

Introduction

A colour palette is a selected set of colours used in a chart, dashboard, map, or other visualization. Effective colour-palette design is important because colours influence how readers interpret information. A poorly designed palette can make a chart confusing, misleading, or difficult to read.

An effective colour palette should communicate information clearly, maintain consistency, and remain accessible to different users. Three important principles are using a limited number of colours, maintaining consistent colour meanings, and ensuring accessibility.

1. Use a Limited Number of Colours

The first principle is to use only the number of colours necessary to communicate the information. Using too many colours can make a chart visually crowded and difficult to interpret.

When a chart contains many unnecessary colours, readers may spend more time understanding the colour differences than understanding the data itself. A limited palette creates a clean appearance and helps emphasize important information.

For categorical data, a small set of clearly distinguishable colours is usually sufficient. For quantitative data, an ordered sequence of shades is often more appropriate.

Example:

Suppose a college wants to compare student enrolment across three departments: Computer Science, Mathematics, and Physics. The chart can use blue, green, and orange to represent the three departments.

Using three distinct colours makes the chart easy to understand. Using ten unrelated colours for the same three departments would add unnecessary complexity.

Importance:

- Improves chart readability.
- Reduces visual confusion.
- Helps readers focus on the data.
- Creates a simple and professional appearance.

Therefore, the number of colours should be appropriate to the number of categories and the purpose of the visualization.

2. Maintain Consistent Colour Meaning Across Charts

The second principle is consistency. A colour should represent the same category or concept throughout related charts and dashboards.

When the same colour is used for different meanings in different charts, readers may become confused. Consistent colour mapping allows users to recognize categories quickly without having to relearn the legend each time.

This principle is especially important in dashboards, reports, and presentations containing several related visualizations.

Example:

Consider a company dashboard that displays the sales performance of three products: Product A, Product B, and Product C.

The colours are assigned as follows:

- Blue represents Product A.
- Green represents Product B.
- Orange represents Product C.

If blue represents Product A in the first chart, it should continue to represent Product A in the second and third charts. Changing Product A to orange in another chart could mislead readers.

Consistency is also useful when colours communicate a status. For example, a dashboard might consistently use red for values requiring attention and green for values meeting a target, provided those meanings are clearly explained.

Importance:

- Makes related charts easier to compare.

- Reduces the need to repeatedly study legends.
- Improves understanding across dashboards.
- Prevents confusion caused by changing colour meanings.
- Creates a consistent visual language.

Therefore, colours should be assigned meaningful roles and reused consistently throughout related visualizations.

3. Ensure Accessibility and Colour-Vision Compatibility

The third principle is accessibility. A colour palette should be understandable to as many readers as possible, including people with colour-vision deficiencies.

Some people have difficulty distinguishing certain colours, particularly combinations involving red and green. If a chart depends entirely on these colours, some readers may struggle to interpret the information correctly.

To improve accessibility, designers should choose colours with sufficient contrast and avoid relying only on colour to communicate important differences.

Labels, symbols, patterns, line styles, and direct annotations can be used alongside colour.

Example:

Suppose a chart compares completed and pending assignments. If completed assignments are shown in green and pending assignments in red, some users may have difficulty distinguishing the two categories.

The chart can be improved by adding labels such as "Completed" and "Pending" and using different symbols or patterns. The colours should also be selected to provide sufficient contrast.

For a temperature map, designers can choose a perceptually ordered palette and include a clearly labelled colour bar so that users can interpret the values.

Importance:

- Makes charts more inclusive.
- Helps people with colour-vision deficiencies.
- Improves readability in different viewing conditions.
- Reduces the chance of misunderstanding.
- Ensures that information remains clear when colours are difficult to distinguish.

Additional Considerations for Colour-Palette Design

In addition to the three main principles, designers should consider the following points:

Meaningful colour selection: Colours should suit the type of information being represented. For example, a sequential palette can show increasing rainfall, while distinct colours can identify different departments.

Sufficient contrast: Text, lines, and data markers should stand out clearly from the background.

Clear legends: A legend or colour bar should explain the meaning of each colour or shade.

Avoid misleading scales: Colour intensity and ordering should accurately reflect the underlying data. A colour scale should not exaggerate small differences or conceal important variations.

Conclusion

Effective colour-palette design improves the clarity, accuracy, and accessibility of data visualizations. Using a limited number of colours prevents visual clutter, maintaining consistent colour meanings helps readers compare related charts, and ensuring accessibility allows more people to understand the information. By following these principles, designers can create charts that communicate data effectively and avoid unnecessary confusion.

Question 4 *[Coordinate Systems and Axes — Cartesian Coordinates]*

A fitness tracker records Steps Walked and Calories Burned for each day of a month. Design a Cartesian-coordinate chart to visualize the relationship between the two, specifying the axes, and explain one insight the chart could reveal.

Cartesian coordinates are widely used in data visualization to represent numerical values using two perpendicular axes: the horizontal X-axis and the vertical Y-axis. They are suitable for displaying relationships between two quantitative variables.

A fitness tracker records the number of steps walked and calories burned each day. Visualizing these measurements can help users understand how their daily walking activity relates to energy expenditure.

A scatter plot using Cartesian coordinates is an appropriate choice because it allows each day's measurements to be displayed as an individual point.

Design of the Cartesian-Coordinate Chart

The chart should represent the relationship between the number of steps walked and the calories burned for each day of the month.

1. X-axis: Steps Walked

The horizontal X-axis represents the total number of steps walked each day.

The axis can begin at zero and increase in regular intervals, such as 2,000, 4,000, 6,000, 8,000, and 10,000 steps.

Each value indicates the daily walking activity recorded by the fitness tracker.

2. Y-axis: Calories Burned

The vertical Y-axis represents the number of calories burned each day.

The scale can use suitable intervals, such as 100, 200, 300, 400, and 500 calories, depending on the recorded data and what the tracker measures.

For accurate interpretation, the chart should specify whether the measurements refer to active calories or total daily energy expenditure.

3. Chart Type: Scatter Plot

A scatter plot is suitable because both steps walked and calories burned are numerical variables.

Each point represents one day of the month. The point's horizontal position indicates the number of steps walked, while its vertical position indicates the number of calories burned.

For example, suppose the tracker records the following measurements:

Day	Steps Walked	Calories Burned
Day 1	2,000	100
Day 2	4,000	180
Day 3	6,000	250
Day 4	8,000	330
Day 5	10,000	400

These are illustrative values used to explain the chart design.

For Day 1, the point would be positioned at 2,000 on the X-axis and 100 on the Y-axis. For Day 5, the point would be positioned at 10,000 on the X-axis and 400 on the Y-axis.

The remaining days can be plotted using the same method.

4. Chart Title and Labels

The chart should include the following details:

- **Title:** Relationship Between Steps Walked and Calories Burned.
- **X-axis label:** Number of Steps Walked per Day.
- **Y-axis label:** Calories Burned per Day.
- **Data points:** One point for each day of the month.
- **Optional trend line:** A line showing the general direction of the relationship.

A trend line can help readers identify whether the relationship between steps and calories burned is positive,

negative, or weak.

Why Cartesian Coordinates Are Suitable

Cartesian coordinates are appropriate because both steps walked and calories burned are quantitative variables.

The horizontal and vertical axes make it easy to compare values and observe relationships. Each observation can be located precisely using its X and Y values.

The scatter plot can also display many daily observations in a single chart, making it possible to identify patterns across the month.

Unlike a polar-coordinate chart, which is particularly useful for circular measurements such as wind direction, a Cartesian chart is more appropriate for examining relationships between two numerical variables that do not naturally form a circular pattern.

One Important Insight the Chart Could Reveal

A positive relationship between steps walked and calories burned.

The scatter plot may show that days with more steps generally correspond to higher calorie expenditure.

For example, the illustrative data show that the number of steps increases from 2,000 to 10,000 while calories burned increase from 100 to 400. If the actual data display a similar pattern, the points would tend to rise from the lower-left region of the chart toward the upper-right region.

This pattern suggests a positive association between walking activity and calories burned.

However, the chart does not prove that steps alone determine calorie expenditure. Other factors, including body weight, walking speed, terrain, exercise intensity, and metabolism, can influence the number of calories burned.

The scatter plot may also reveal unusual observations. For example, a day with relatively few steps but high calorie expenditure might indicate that the person performed another form of exercise.

Conclusion

A Cartesian-coordinate scatter plot is an effective method for visualizing the relationship between steps walked and calories burned. The X-axis represents daily steps, and the Y-axis represents daily calories burned. Each point corresponds to one day of the month. The chart can reveal a positive association between walking activity and energy expenditure, identify unusual observations, and help users better understand their fitness patterns.

Question 5 *[Mapping Data onto Aesthetics — Types of Data]*

A dataset records, for each patient, Age (years), Blood Group (A/B/AB/O), Satisfaction (Low/Medium/High), and City (name). Classify each variable by data type (nominal, ordinal, discrete, or continuous), and explain your reasoning for one variable that could reasonably be interpreted two ways.

Introduction

Data classification is an important step in data analysis and visualization. Before selecting a chart or mapping data onto visual properties, such as colour, position, size, or shape, it is necessary to understand the type of each variable.

Different variables contain different kinds of information. Some variables describe categories, while others represent numerical measurements or counts.

The given patient dataset contains four variables: Age, Blood Group, Satisfaction, and City. These variables can be classified as nominal, ordinal, discrete, or continuous according to their characteristics and how they are recorded.

1. Age — Continuous Data

Age represents the amount of time a person has lived. It is a quantitative variable because it can be measured numerically.

Age is generally classified as continuous when it is measured precisely. It can take fractional values, depending on the measurement method.

For example, a patient's age could be recorded as 25.5 years, 25.75 years, or 25 years and 6 months.

Because age can take many possible values within a range, it is generally treated as a continuous variable when precise measurements are available.

Example: A patient is recorded as 32.5 years old.

Reason: Age represents a measurable quantity and can take fractional values.

Suitable visualization: A histogram can show the distribution of patient ages. A box plot can summarize the median, spread, and potential outliers.

2. Blood Group — Nominal Data

Blood group is a categorical variable that identifies a patient's blood type. The common ABO blood groups are A, B, AB, and O.

These categories do not have a natural numerical order. For example, blood group A is not greater than blood group B, and blood group O is not less than blood group AB.

Therefore, blood group is classified as nominal data.

Nominal variables are used to identify categories or labels rather than numerical quantities or rankings.

Example: A patient has blood group B.

Reason: Blood groups identify categories that have no natural order.

Suitable visualization: A bar chart can display the number of patients in each blood-group category. Different colours can distinguish the categories, provided a clear legend is included when necessary.

3. Satisfaction — Ordinal Data

Satisfaction is a categorical variable that describes a patient's level of satisfaction with a service or experience.

The given categories are Low, Medium, and High. These categories have a meaningful order:

Low < Medium < High

However, the numerical difference between the categories is not necessarily equal. The difference between Low and Medium satisfaction may not be equivalent to the difference between Medium and High satisfaction.

For this reason, satisfaction is classified as ordinal data.

Ordinal data consists of categories that can be ranked according to a meaningful order, even when the differences between ranks cannot be measured precisely.

Example: A patient reports a High level of satisfaction.

Reason: The categories have a natural ranking, but the intervals between them are not necessarily equal.

Suitable visualization: An ordered bar chart can display the number of patients reporting Low, Medium, and High satisfaction. The categories should appear in their natural order rather than alphabetical order.

4. City — Nominal Data

City represents the name of the city where a patient lives or receives treatment, depending on how the dataset defines the variable.

Examples of city names could include Chennai, Bengaluru, Hyderabad, and Mumbai.

City names are categorical labels. They do not have a natural ranking, and one city is not inherently greater or smaller than another.

Therefore, City is classified as nominal data.

Example: A patient is associated with Chennai.

Reason: City names identify categories without a meaningful natural order.

Suitable visualization: A bar chart can compare the number of patients associated with each city. If geographical location is important, a map may also be appropriate, provided reliable location information is available.

Summary of Data Types

Variable	Data Type	Explanation	Example
Age	Continuous	Can be measured precisely and can include fractional values.	32.5 years
Blood Group	Nominal	Categories have no natural order.	A, B, AB, O
Satisfaction	Ordinal	Categories follow a meaningful order.	Low, Medium, High
City	Nominal	City names are categories without a natural ranking.	Chennai, Mumbai

Variable That Can Be Interpreted in Two Ways: Age

Age is a good example of a variable that can reasonably be interpreted in two ways depending on how it is recorded.

Interpretation 1: Age as Continuous Data

Age is continuous when it is measured precisely. A person's age can be expressed in years, months, days, or smaller units of time.

For example:

- 18.5 years
- 25.75 years
- 40.25 years

These values demonstrate that age can take fractional values within a given range.

In this situation, age is treated as continuous because it represents a measurement rather than a simple count.

Interpretation 2: Age as Discrete Data

Age can be treated as discrete when the dataset records only completed years.

For example:

- 18 years
- 25 years
- 40 years

If the dataset stores age only as whole numbers, such as 18, 25, and 40, the recorded variable takes integer values.

In practical data analysis, age recorded in completed years is often treated as a discrete variable because the recorded values increase in whole-year steps.

However, the underlying quantity—time lived—is still continuous. It is the recording method that makes the observed values discrete.

Why the Classification Matters

Understanding how age is recorded helps analysts select appropriate summaries and visualizations.

If age is recorded precisely, a histogram can display the distribution of continuous measurements. If age is grouped into completed years, an ordered frequency table or bar chart may also be appropriate, depending on the purpose and size of the dataset.

Similarly, the other variables require appropriate visual treatments. Blood group and city can be represented using distinct colours because they are nominal categories. Satisfaction should preserve the order of Low, Medium, and High because it is ordinal.

Importance of Classifying Variables Correctly

Correctly classifying data types helps analysts choose appropriate charts and avoid misleading interpretations.

1. Appropriate chart selection: Continuous numerical variables can often be visualized using histograms, while categorical variables can be represented using bar charts.

2. Correct use of colour: Distinct colours are suitable for nominal categories, whereas ordered colour scales may be

suitable for quantitative measurements.

3. Accurate interpretation: Recognizing satisfaction as ordinal prevents analysts from assuming that the differences between Low, Medium, and High are necessarily equal.

4. Better data analysis: Correct classification helps determine which statistical summaries and analytical methods are appropriate.

5. Clear communication: A well-designed visualization presents the data in a way that reflects the actual meaning of each variable.

Conclusion

The patient dataset contains variables with different data types. Age is generally continuous when measured precisely, blood group is nominal because its categories have no natural order, satisfaction is ordinal because its categories have a meaningful ranking, and city is nominal because city names are categorical labels.

Age can also be treated as discrete when recorded only in completed years. This example demonstrates that data classification can depend on how a variable is measured and recorded. Correctly identifying data types helps analysts select suitable visualization techniques and communicate information clearly and accurately.